Eric Walker

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EDUCATION

University of Colorado Boulder **August 2023 - May 2027**

Bachelor of Science in Electrical and Computer Engineering | Minor in Computer Science

- **GPA:** 3.61
- **Relevant Courses:** Circuits and Electronics; Circuits as Systems; Microelectronics; Digital Logic; Embedded Systems; Embedded Software; Database Systems; Computer Systems; Data Structures; Operating Systems

SKILLS & INTERESTS

- **Programming Languages:** C, C++, Python, SQL, Verilog, MATLAB
- **Hardware & Tools:** Oscilloscopes, Multimeters, Soldering, PCB design (Altium), 3D printing, SPICE, CAD
- **Interests**: Skiing; Running; Cooking; Traveling; Biking

TECHNICAL PROJECTS

Wi-Fi-Controlled Discrete Circuit Robot: **March 2026**

- Designed and built a closed-loop robotic control system powered by an Arduino Nano Every, integrating dual speed sensors and hardware compensator circuits for precise, programmed movement control.
- Constructed discrete motor driver circuitry using MOSFETs and BJTs for bidirectional movement, utilizing encoder pulse tracking to execute exact rotational and linear movements.
- Integrated an ESP-01 module utilizing ESP-NOW protocol for low-latency wireless communication, enabling remote control through both a mobile web interface and a custom Raspberry Pi Pico joystick.
- *Skills Used: Circuit Design, SIMetrix (Circuit Simulation), Arduino IDE Programming, Closed Loop Control Systems*

Sumo Robot Embedded System: **September 2025**

- Collaborated in a 3-person team to design and build a remotely operated sumo robot using Raspberry Pi Pico microcontrollers, DC motors, voltage regulators, a custom PCB, and 3D-printed mechanical assemblies.
- Developed a custom Raspberry Pi Pico-based wireless controller utilizing IR communication for real-time robot control during competition.
- *Skills Used: Python, PCB Design (Altium), IR / RF Communication, 3D Printing*

STM32 BlackJack Game (Embedded Programming): **August 2025**

- Designed and implemented a functional Blackjack game in C on the STM32F429I Discovery board, utilizing the onboard LCD and touchscreen for real-time game state visualization and user interaction.
- Implemented interrupt-driven touchscreen input handling to process player actions (hit/stand) and render card graphics dynamically.
- Implemented low-level peripheral control, system timing, and hardware register manipulation to manage game logic and display rendering.
- *Skills Used: C, STM32CubeIDE, GPIO / Interrupt Programming*

TECHNICAL WORK EXPERIENCE

Colorado Space Grant, CARMA Tech Team: **May 2026 - Present**

- Built, calibrated, and deployed NASA STELLA open-source spectrometers to capture visible and near-infrared (NIR) irradiance telemetry for research into plant heat-stress detection.
- Designed and constructed sensor housing for long-term soil off-gassing and environmental monitoring, including custom enclosures and PCBs to be used in conjunction with STELLAS for environmental research.

ADDITIONAL WORK EXPERIENCE

- **Produce Clerk** | King Soopers (Erie, CO) | *May 2025 – April 2026*
- **Food Court Clerk** | Costco Wholesale (Longmont, CO) | *May 2024 – August 2024*
- **Courtesy Clerk** | Sprouts Farmers Market (Lafayette, CO) | *May 2022 – November 2022*